

Bern EQ

A three-band broadcast channel equaliser with transformer-coupled input and output stages.



Overview

Bern EQ is the channel equaliser in the spirit of the classic Swiss broadcast consoles, and the companion of the Bern 16 summing mixer. It is musical, very open and very clean: a LOW section that is either a gentle shelf or a high-pass filter, a sweepable MID presence bell, and a smooth HIGH shelf for air.

The EQ sits between two amplifiers, as on the desk. The input amplifier has about 14 dB of headroom above 0 dBFS, so it is transparent at normal levels; push it with INPUT and a little even-harmonic warmth comes in, then a soft, round limit. The output amplifier feeds a transformer whose core thickens the lows first when it is driven.

The bands are carefully matched to their analogue curves, so the bell and the high shelf keep their true shape right up to the top of the audio range. Every control glides, so sweeping and switching never click.

Reach for it for

- Clean, broad tone shaping of vocals, guitars, piano and buses
- Adding glassy air with the high shelf without harshness
- Clearing rumble with the high-pass before anything else

- Presence and focus with the MID bell
- A touch of console warmth by driving the input
- Matching channels with Bern 16 for a whole-desk sound

Controls

The header

PRESETS	Opens the preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
EQ IN	Switches the three bands in or out (in by default). The amplifiers, the transformer and the OUT level stay in either way, as on the desk, so you hear the EQ alone when you compare.
DRIFT	How much the supply voltage wanders, 0 to 100% (20% by default). Its value shows beside it while you turn it. See Drift below.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper that holds the peaks at -0.3 dBFS (no latency), or a lookahead limiter that holds them at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. A new instance opens in L.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Response window

Shows the EQ curve from 20 Hz to 20 kHz, plus and minus 18 dB. Each band is drawn faintly in its own colour (LOW blue, or yellow for the high-pass, MID green, HIGH red) with the total curve as the bright line, and a dot marks each band's frequency. The window reads EQ OUT and the curve goes flat when EQ IN is off.

INPUT

TRIM	Level into the input amplifier, -18 to +18 dB (0 dB by default). Turn it up to drive the amplifier for warmth.
WARMTH lamp	Glowes amber as the input amplifier works harder. Off means it is clean.
Signal flow	IN AMP, EQ, IRON, OUT: the order the signal takes. EQ dims while EQ IN is off.

LOW

The LOW section is either a shelf or a high-pass filter. Pressing a SHELF key picks the shelf; pressing a HIGH-PASS key picks the filter. Switching between them crossfades without a click.

GAIN	Boost or cut for the low shelf, -15 to +15 dB (0 dB by default). It does nothing in high-pass mode, and its cap fades to show that.
SHELF 60 / 100	A gentle low shelf at 60 or 100 Hz (100 Hz by default; the shelf is the default mode).
HIGH-PASS 40 / 80 / 120	A clean 12 dB per octave high-pass at 40, 80 or 120 Hz, in place of the shelf.

MID PRESENCE

GAIN	Boost or cut, -15 to +15 dB (0 dB by default).
FREQ	The centre of the bell, 150 Hz to 7 kHz (1.2 kHz by default).
BROAD / NARROW (Q)	The width of the bell. BROAD (the default) is a musical, moderately wide curve; NARROW is about two and a half times tighter, for more focused work.

HIGH

GAIN	Boost or cut, -15 to +15 dB (0 dB by default).
HINGE 10 / 15 / 20 kHz	Where the high shelf is half-way to its full gain (15 kHz by default). The slope is gentle, with no bump, so a boost reaches well down into the upper mids: 10 kHz is the most obvious, 20 kHz the most airy.

OUT

LEVEL	Output level after the output amplifier and transformer, -18 to +18 dB (0 dB by default).
Meter	A 16-segment output meter, about -30 to 0 dBFS: green, amber near the top and red at the top two segments.

Drift

DRIFT lets the supply voltage wander slowly and smoothly, as on a real console. As it moves, the band gains ease by up to 2.5% and the band frequencies by up to 0.6% (at 100%), and the amplifiers' headroom rises and falls by up to 2 dB, so a hot signal softens a little more when the rail sags. At 0% the EQ stays exactly where you set it.

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click to reset a knob to its default.

Presets

Presets set INPUT, the three bands and the OUT level, and switch EQ IN on. Init also resets DRIFT and the output stage; the others leave them as they are.

Init	The defaults: everything flat, 100 Hz shelf, 1.2 kHz BROAD bell, 15 kHz hinge.
Vocal Air	High-pass at 80 Hz, +1.5 dB at 3.2 kHz and +5 dB of air at 15 kHz; OUT -1 dB.
Broadcast Voice	INPUT +3 dB for a little warmth, high-pass at 120 Hz, +3 dB at 2.5 kHz, +2 dB at 10 kHz; OUT -4 dB.

Kick Weight	+6 dB at 60 Hz, -3.5 dB of box at 320 Hz, a little top at 10 kHz.
Guitar Presence	High-pass at 80 Hz, +3.5 dB at 2.2 kHz, +1.5 dB at 10 kHz.
Bass Clean-Up	High-pass at 40 Hz, -3 dB at 250 Hz and -2 dB at 10 kHz to tidy the top.
Mix Bus Sheen	+1 dB at 60 Hz, a hair less at 3 kHz and +3 dB of air at 20 kHz.
Telephone	INPUT +6 dB, high-pass at 120 Hz, a big NARROW +10 dB at 1.5 kHz and -15 dB at 10 kHz; OUT -8 dB.
De-Mud	-5 dB at 300 Hz with a slight low trim and +1.5 dB at 15 kHz.
Piano Open	A little less low end, -2 dB at 600 Hz and +3 dB at 15 kHz.
Warm Drive	INPUT +4 dB into the amplifier, +2.5 dB at 60 Hz, +1 dB at 1.8 kHz, +2 dB at 20 kHz; OUT -4 dB to match.

Tips

- For a vocal, start with HIGH-PASS 80 and add HIGH gain at 15 kHz before reaching for the MID bell.
- The high shelf is very gentle, so you can boost more than you would expect before it gets harsh.
- To hear the amplifiers, turn TRIM up until the WARMTH lamp glows, then bring LEVEL down by the same amount.
- Use EQ IN to compare: the amplifiers stay in circuit, so you hear only what the bands are doing.
- Set DRIFT to 0 for a static, repeatable EQ; leave a little in for a less clinical, hardware-like feel.

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